

User Persona: Downstream Health Data Scientist & Epidemiologist

DATA QUALITY → DECISION CONFIDENCE

Project: NYU SHC Predictive Analytics & Data Cleaning Pipeline

NYU SHC • USER PERSONA



Dr. Emily Chen

Lead Health Data
Scientist /
Biostatistician

BACKGROUND

6+ years in biostatistics & epidemiological modeling
Tools: Python (pandas, scikit-learn),
R, SQL, Tableau

CORE GOALS

- Build real-time outbreak risk models for flu & COVID-19.
- Identify high-risk cohorts, including final-exam mental-health stress.
- Deliver weekly health-trend reports to university leadership.

FROM VALIDATED DATA TO CAMPUS-LEVEL ACTION



Pain Points

Dirty & inconsistent data: missing values, mismatched timestamps, duplicate check-in IDs.
Privacy anxiety: PHI exposure risk during HIPAA/FERPA collaboration.
No stable dictionary: survey codes and diagnosis fields shift by semester.

70%
TIME CLEANING



Key Decisions to Make

SHE NEEDS TRUSTWORTHY EVIDENCE TO ACT

- “Which clinical features are clean enough for our early-warning disease model?”
- “Where should clinic staff and mental-health counselors be allocated next month?”
- “Is this a true campus trend—or a duplicate-entry artifact?”



How Our Pipeline Helps

Standardized Feature Store: pre-cleaned tables with validated schemas.
Automated De-identification: PHI stripped for compliant collaboration.
Hours, not weeks: train and deploy predictive models faster.

80%
TIME SAVED